

News Encoder → Survey Expectation Pipeline

Technical Documentation

FinBERT · Random Forest · Walk-Forward Evaluation

This document describes the complete data pipeline for mapping aggregated WSJ financial-news embeddings to survey-based macroeconomic expectations (dividend and earnings growth). It covers all pipeline stages, configuration parameters, design choices, and evaluation methodology.

1. Pipeline Overview

The pipeline transforms raw news headlines into quarterly survey forecasts through eight sequential stages. Each stage is self-contained and its output is passed directly to the next.

Step	Stage	Description
1	Environment Setup	Install dependencies, set seeds, detect GPU for encoder.
2	Load News Corpus	Read WSJ parquet file (1965–2014). Expected columns: date, text. Filtered to articles with non-null headline text.
3	Load Survey Data	Two quarterly series loaded from Excel: dividend growth expectations and earnings growth expectations. Each row contains wave_date and median_forecast.
4	FinBERT Encoding	Frozen ProsusAI/finbert produces 768-dim CLS embeddings per article. A second frozen classification head produces P(neutral) for filtering. Embeddings cached to disk keyed by corpus hash.
5	Sentiment Filtering	Articles with P(neutral) above a tuned threshold are dropped before aggregation. Threshold is jointly optimised with RF hyperparameters via LOO-CV on the initial training window.
6	Wave Aggregation	Remaining articles in each 30-day wave window are collapsed to a single vector via recency-weighted mean (and optionally appended within-wave std for earnings). Output: one vector per survey wave.
7	RF Training & Eval	Hyperparameters tuned once on the initial MIN_TRAIN_WAVES window. Walk-forward expanding-window evaluation produces honest one-step-ahead predictions. Final model refit on all waves for extrapolation.
8	Explainability	Vocabulary saliency word clouds, attention x gradient attribution, feature importance factor interpretation, Cook's D wave influence, and temporal saliency plots.

2. Data Loading

2.1 News Corpus

The news corpus is a Parquet file of Wall Street Journal headlines spanning 1965–2014, compressed with zstd. The pipeline expects two columns:

- **date** — publication date, parsed as datetime64.
- **text** — headline string.

Parameter	Default	Type	Description
NEWS_PARQUET	wsj_headlines_1965_2014_zstd.parquet	Path	Path to parquet file.
AGG_WINDOW_DAYS	30	int	Lookback window in days before each wave closing date.

2.2 Survey Data

Two survey series are loaded from Excel files. Each is processed identically by `run_pipeline()` — the series name determines per-series settings (variance features, training window size).

- **dividend_growth** — 49 waves with news coverage. Short series; walk-forward evaluation uses 9 held-out steps from a 40-wave initial window.
- **earnings_growth** — 157 waves with news coverage. Primary series; 117 walk-forward evaluation steps.

Note: The survey wave date is the forecast closing date. News within the preceding AGG_WINDOW_DAYS days is treated as the information set available to respondents.

3. FinBERT Encoding

3.1 Model objects

Two objects are loaded from the same ProsusAI/finbert checkpoint and kept frozen throughout. Neither is fine-tuned.

Object	Class	Output	Purpose
model	AutoModel	768-dim CLS vector	Article embedding for RF input.
sentiment_model	AutoModelForSequenceClassification	3-class logits	P(neutral) for sentiment filter.

3.2 Encoding procedure

Articles are batched through model in groups of `ENCODE_BATCH`. For each article the [CLS] token from the last hidden state (index 0) is taken as the document representation. Truncation at `MAX_TOKEN_LEN=512` tokens applies to long headlines. Embeddings are cached to disk as a .npz array keyed by corpus size and date range hash to avoid re-encoding on subsequent runs.

Parameter	Default	Type	Description
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ENCODER_NAME	ProsusAI/finbert	str	HuggingFace model ID. Both objects load from this checkpoint.
MAX_TOKEN_LEN	512	int	BERT context window. Headlines rarely exceed 64 tokens in practice.
ENCODE_BATCH	32	int	Batch size for encoder forward passes. Reduce if OOM on GPU.

3.3 FinBERT label order

ProsusAI/finbert uses the label ordering: 0 = positive, 1 = negative, 2 = neutral. The pipeline reads $P(\text{neutral})$ as $\text{softmax}(\text{logits})[:, 2]$. This ordering is hardcoded as `FINBERT_NEUTRAL_IDX = 2` in the config.

4. Wave Aggregation

4.1 Sentiment filtering

Before aggregating, each article's $P(\text{neutral})$ score from `sentiment_model` is compared against a threshold. Articles exceeding the threshold are dropped. The intuition is that articles FinBERT classifies as highly neutral carry little directional information about expectation formation — they are boilerplate market summaries rather than substantive macro news.

If the filter would remove all articles in a wave, the single least-neutral article is retained to avoid a missing wave. This ensures no survey wave is dropped purely due to an aggressive threshold.

Parameter	Default	Grid	Description
NEUTRAL_THRESHOLD	0.90	[0.75, 0.85, 0.90, 0.95, 1.01]	$P(\text{neutral})$ cutoff. 1.01 = no filtering. Tuned via LOO-CV.
SENTIMENT_BATCH	64	—	Batch size for <code>sentiment_model</code> forward passes.

4.2 Recency-weighted mean

After filtering, articles are aggregated to a single 768-dim wave vector using an exponentially decaying recency weight:

$$w_i = 0.5^{(\text{days_back}_i / \text{HALFLIFE})} \quad z_{\text{wave}} = \text{sum}(w_i * \text{emb}_i) / \text{sum}(w_i)$$

Articles closer to the wave closing date receive higher weight. Dividing by the sum of weights normalises for the variable number of surviving articles so the wave vector is always on the same scale.

Parameter	Default	Type	Description
REGENCY_WEIGHTING	exponential	str	"uniform" or "exponential". Exponential is default.
REGENCY_HALFLIFE_DAYS	7	int	Half-life in days for exponential decay.

4.3 Within-wave variance features (earnings only)

For `earnings_growth` only, the recency-weighted standard deviation of article embeddings within each wave is computed and appended to the mean vector, producing a 1536-dim feature vector. This captures within-quarter disagreement across articles — a proxy for interpretive heterogeneity in the information environment.

Variance features are disabled for `dividend_growth` because the series has only 49 waves with a median of fewer than 10 articles per wave — too few to estimate within-wave variance reliably.

Series	USE_VARIANCE_FEATURES	Feature dim	Rationale
dividend_growth	False	768	49 waves, low article count per wave.
earnings_growth	True	1536	157 waves, regime sensitivity in RMSE plot.

5. Random Forest Model

5.1 Model choice

RandomForestRegressor is fit directly on raw 768-dim (or 1536-dim) embeddings without PCA pre-compression. Trees regularise high-dimensional input natively through `max_features` subsampling at each split, so PCA would discard signal without providing regularisation benefit.

Extra Trees were also evaluated. Random Forest was preferred because at $N < 150$ quarterly waves, optimised splits (RF) outperform fully random splits (Extra Trees) when `min_samples_leaf` is tuned aggressively — the optimised splits find the sparse relevant dimensions more reliably.

5.2 Hyperparameter tuning

Hyperparameters are tuned once via leave-one-out cross-validation on the first `MIN_TRAIN_WAVES` waves. The search is joint over three axes: the `neutral_threshold` (aggregation), `min_samples_leaf`, and `max_features`. LOO-CV is used rather than k-fold because at quarterly N , exact LOO avoids the variance of random fold assignments and correctly treats each observation as a single held-out data point.

Parameter	Grid	Notes
<code>neutral_threshold</code>	[0.75, 0.85, 0.90, 0.95, 1.01]	Jointly tuned with RF params.
<code>min_samples_leaf</code>	[1, 2, 3, 5, 8, 13, 20, 30]	Primary regularisation lever.
<code>max_features</code>	["sqrt", "log2", 0.05, 0.10]	Feature subsampling per split.
<code>n_estimators</code>	500 (fixed)	Not tuned; 500 is sufficient.

Note: Hyperparameters are fixed after the initial tuning step and held constant throughout the walk-forward loop. Re-tuning at each step would leak information about which hyperparameters generalise across the full sample.

5.3 Sample weights

Each wave is weighted in RF training by $\log(1 + n_{\text{after_filter}})$, normalised to unit mean. Waves with more surviving articles after sentiment filtering are treated as more reliable estimates of the true quarter sentiment and receive higher weight.

```
sw_i = log(1 + n_articles_after_filter_i) sw = sw / mean(sw)
```

6. Walk-Forward Evaluation

6.1 Motivation

A fixed train/test split evaluated repeatedly during model selection causes the held-out window to leak into model selection decisions even without direct training on it. Walk-forward evaluation eliminates this by producing one genuinely out-of-sample prediction per evaluation step.

Standard k-fold cross-validation is inappropriate for time series because random fold assignment trains on future data to predict the past within each fold. The walk-forward scheme respects temporal ordering throughout.

6.2 Scheme

Let N be the total number of waves with news coverage and $M = \text{MIN_TRAIN_WAVES}$. The walk-forward loop runs $N - M$ steps:

Step	Train on	Predict	AR(1) baseline
1	waves 1...M	wave M+1	$y[M]$
2	waves 1...M+1	wave M+2	$y[M+1]$
■	■	■	■
N-M	waves 1...N-1	wave N	$y[N-1]$

Parameter	dividend_growth	earnings_growth	Notes
MIN_TRAIN_WAVES	40	40	Initial tuning + first train window.
Walk-forward steps	9	117	N minus MIN_TRAIN_WAVES.
Rolling RMSE window	8	12	Auto-selected based on step count.

6.3 Reported metrics

Metric	Formula	Interpretation
WF RMSE	$\sqrt{\text{mean}((y - \hat{y})^2)}$	Primary performance metric.
AR(1) RMSE	$\sqrt{\text{mean}((y - y[t-1])^2)}$	Naive persistence benchmark.
WF R^2	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$ (WF window only)	Proportion of variance explained.
Dir. accuracy	$\text{mean}(\text{sign}(\Delta y) == \text{sign}(\Delta \hat{y}))$	Directional prediction accuracy.
Rolling RMSE	8 or 12-step rolling mean of sq. errors	Regime-level performance.

6.4 Final model

After walk-forward evaluation, the model is refit on all N waves at the tuned hyperparameters. This final model is used for explainability analysis and historical extrapolation. In-sample R^2 on the final model is expected to be high by construction and should not be reported as a performance metric.

7. Explainability

Five explainability analyses are run on the final model. All use the RF importance-weighted embedding direction reconstructed as: $w_emb = feature_importances * sign(corr(Z_dim, y))$.

Method	What it shows	Key output
Vocabulary saliency word cloud	Every FinBERT vocabulary token embedded in the headline, projected to an F_{emb} sized proportional to the dot product of the token embedding and the headline embedding	Batch of Polish projected to an F_{emb} sized proportional to the dot product of the token embedding and the headline embedding
Attention x gradient attribution	For a selected headline: attention weights (CLS token) of pre-trained attribution of predicted headline via CoVe	Batch of Polish projected to an F_{emb} sized proportional to the dot product of the token embedding and the headline embedding
Feature importance factor interpretation	Top-K embedding dimensions by RF feature importance	Polish tokens (tokens) of pre-trained attribution of predicted headline via CoVe
Wave influence (Cook's D proxy)	LOO re-fits of RF at tuned hyperparameters. Shap for an influence scores	Top 10 flagged waves Top 3 headlines
Temporal saliency	Recency-weighted mean dot product of article embeddings with series comparison for every wave including pre-survey	Batch of Polish projected to an F_{emb} sized proportional to the dot product of the token embedding and the headline embedding

8. Full Configuration Reference

All parameters are defined in the Configuration cell (cell 5) of the notebook. TODO markers indicate values that require manual review before running.

Parameter	Default	TODO?	Description
DATA_DIR	./data	Yes	Root directory for parquet and Excel files.
CACHE_DIR	./cache	No	Embedding cache location.
OUTPUT_DIR	./output	No	Predictions, plots, and saved models.
ENCODER_NAME	ProsusAI/finbert	No	HuggingFace model ID for both model objects.
MAX_TOKEN_LEN	512	No	Tokeniser truncation length.
ENCODE_BATCH	32	No	Encoder batch size. Reduce if GPU OOM.
FINBERT_NEUTRAL_IDX	2	No	Index of neutral class in FinBERT output.
AGG_WINDOW_DAYS	30	No	Days before wave date included in window.
REGENCY_WEIGHTING	exponential	No	"uniform" or "exponential".
REGENCY_HALFLIFE_DAYS	7	No	Exponential decay half-life in days.
NEUTRAL_THRESHOLD	0.90	No	Default threshold; overridden by LOO-CV grid.
NEUTRAL_THRESHOLD_GRID	[0.75–1.01]	No	LOO-CV search grid. 1.01 = no filtering.
SENTIMENT_BATCH	64	No	Batch size for sentiment_model.
USE_VARIANCE_FEATURES	per-series dict	No	True for earnings_growth, False for dividends.
MIN_TRAIN_WAVES_PER_SERIES	40 both	Yes	Adjust if series length changes.
RF_N_ESTIMATORS	500	No	Fixed; not part of LOO-CV search.
RF_MSL_GRID	[1,2,3,5,8,13,20,30]	No	min_samples_leaf search grid.
RF_MAX_FEATURES_GRID	["sqrt","log2",0.05,0.10]	No	max_features search grid.
PCA_COMPONENTS_MAHAL	30	No	PCA dims for Ledoit-Wolf covariance estimate.
SEED	42	No	Random seed for RF and PCA.

9. Key Design Decisions & Rationale

Frozen encoder

FinBERT is never fine-tuned. With $N < 150$ quarterly waves, end-to-end fine-tuning would overfit. The frozen CLS embedding is treated as a fixed feature extractor whose representations were pre-trained on financial text via masked language modelling.

No PCA pre-compression

Unlike Ridge regression, Random Forests regularise high-dimensional input natively through max_features subsampling. PCA before the forest discards signal without providing regularisation benefit, and was found to hurt out-of-sample performance.

Joint threshold + RF tuning

The neutral sentiment threshold is tuned jointly with RF hyperparameters via the same LOO-CV objective. Separating threshold selection from model selection would risk optimising an intermediate step on a different criterion than the final prediction target.

One-time hyperparameter tuning

Hyperparameters are tuned once on the initial training window and fixed for all walk-forward steps. Re-tuning at each step would constitute look-ahead bias since the set of hyperparameters that generalises well across all evaluation steps is information about the full sample.

LOO-CV over k-fold

At quarterly $N = 40\text{--}157$, LOO is exact and avoids the variance of random fold assignments. It is also the natural counterpart to the walk-forward evaluation scheme, which is itself a form of LOO in time.

Sample weights by article count

$\log(1 + n_{\text{articles}})$ sample weights tell the forest that waves with more surviving articles are more reliable estimates of true-quarter sentiment. This is a principled response to the variable- N aggregation problem rather than discarding that information.

Variance features for earnings only

Within-wave embedding std is a proxy for interpretive heterogeneity — disagreement across articles in a quarter. It is enabled for earnings (157 waves, rolling RMSE shows regime sensitivity) and disabled for dividends (49 waves, too few articles per wave to estimate variance).

*For fine-tuning FinBERT on survey labels, see the Optional: Fine-tuning FinBERT section in the notebook (cell 35).
For historical extrapolation setup, see the Historical Extrapolation section (cell 36).*